

Units and Measurement

1.1 Introduction to Measurement and Units

Measurement is a comparison process involving a chosen, internationally accepted reference standard known as a

unit. The outcome of a physical quantity measurement is expressed as a numerical value accompanied by its unit. While there are many physical quantities, only a limited number of units are needed because these quantities are interrelated.

- **Fundamental (Base) Units:** These are units for the basic quantities, such as length, mass, and time.
- **Derived Units:** Units for other physical quantities, which are expressed as combinations of base units (e.g., speed, density).
- **System of Units:** A complete collection of both base and derived units forms a system of units.

1.2 The International System of Units (SI)

Historically, different countries used various systems like CGS (centimetre, gram, second), FPS (foot, pound, second), and MKS (metre, kilogram, second) . The

Système Internationale d'Unités (SI), adopted in 1971 by the Bureau International des Poids et Mesures (BIPM) and revised in November 2018, is the currently accepted international system. Its decimal nature simplifies conversions.

The SI system consists of

seven base units:

1. **Metre (m):** SI unit of length. Defined by taking the fixed numerical value of the speed of light in vacuum (c) as 299,792,458 when expressed in m s^{-1} .
2. **Kilogram (kg):** SI unit of mass. Defined by taking the fixed numerical value of the Planck constant (h) as $6.62607015 \times 10^{-34}$ when expressed in $\text{J s (kg m}^2 \text{ s}^{-1})$.
3. **Second (s):** SI unit of time. Defined by taking the fixed numerical value of the caesium frequency ($\Delta\nu_{\text{Cs}}$) of the caesium-133 atom as 9,192,631,770 when expressed in $\text{Hz (s}^{-1})$.
4. **Ampere (A):** SI unit of electric current. Defined by taking the fixed numerical value of the elementary charge (e) as $1.602176634 \times 10^{-19}$ when expressed in C (A s) .
5. **Kelvin (K):** SI unit of thermodynamic temperature. Defined by taking the fixed numerical value of the Boltzmann constant (k) as 1.380649×10^{-23} when expressed in $\text{J K}^{-1} (\text{kg m}^2 \text{ s}^{-2} \text{ K}^{-1})$.
6. **Mole (mol):** SI unit of amount of substance. One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities (Avogadro constant).

7. **Candela (cd):** SI unit of luminous intensity. Defined by taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz, Kcd, as 683 when expressed in lm W^{-1} (cd sr W^{-1} or $\text{cd sr kg}^{-1} \text{m}^{-2} \text{s}^3$).

Beyond the base units, there are

two supplementary units:

- **Radian (rad):** For plane angle, defined as the ratio of arc length (ds) to the radius (r) ($d\theta = ds/r$ radian).
- **Steradian (sr):** For solid angle, defined as the ratio of the intercepted area (dA) on a spherical surface to the square of its radius (r) ($d\Omega = dA/r^2$ steradian).

Both radian and steradian are

dimensionless quantities.

The textbook also lists

some units retained for general use (though outside SI), including:

- **minute (min):** 60 s
- **hour (h):** 3600 s
- **day (d):** 86400 s
- **litre (L):** $1 \text{ dm}^3 = 10^{-3} \text{ m}^3$
- **tonne (t):** 10^3 kg
- **bar (bar):** $0.1 \text{ MPa} = 10^5 \text{ Pa}$
- **degree ($^\circ$):** $(\pi/180) \text{ rad}$

1.3 Significant Figures

Significant figures (or significant digits) in a measurement indicate its precision. They consist of all reliably known digits plus the first uncertain digit. For example, in 1.62 s, 1 and 6 are reliable, while 2 is uncertain, giving three significant figures. A change in units does not alter the number of significant figures in a measurement.

Rules for Determining the Number of Significant Figures :

1. **All non-zero digits are significant.**
2. **Zeros between two non-zero digits are significant**, regardless of the decimal point's location.
3. **If the number is less than 1, zeros to the right of the decimal point but to the left of the first non-zero digit are *not* significant.** (e.g., in 0.002308, the underlined zeros are not significant).
4. **Terminal or trailing zeros in a number *without* a decimal point are *not* significant.** (e.g., 12300 cm has three significant figures).

5. **Trailing zeros in a number *with* a decimal point are significant.** (e.g., 3.500 or 0.06900 each have four significant figures).
6. **The digit 0 conventionally placed to the left of a decimal for a number less than 1 (e.g., 0.1250) is never significant.** However, zeros at the end of such a number are significant in a measurement.

Scientific Notation ($a \times 10^b$) is recommended to avoid ambiguity, as all zeroes in the base number 'a' are significant. The power of 10 is irrelevant to the determination of significant figures.

Rules for Arithmetic Operations with Significant Figures :

1. **Multiplication or Division:** The final result should retain the same number of significant figures as the original number with the *least* significant figures. (e.g., Density = $4.237 \text{ g} / 2.51 \text{ cm}^3 = 1.69 \text{ g cm}^{-3}$, reported to three significant figures because 2.51 cm^3 has two significant figures, but the example reports to 3 because it says 2.51 is three sig figs. Rechecking the example given: "5.74 g of a substance occupies 1.2 cm^3 . Express its density by keeping the significant figures in view. Answer: There are 3 significant figures in the measured mass whereas there are only 2 significant figures in the measured volume. Hence the density should be expressed to only 2 significant figures. Density = $5.74 / 1.2 \text{ cm}^3 = 4.8 \text{ g cm}^{-3}$ ". So the rule is correct, and the example 1.2 uses it correctly. The example with density = $4.237 \text{ g} / 2.51 \text{ cm}^3 = 1.69 \text{ g cm}^{-3}$ indicates 2.51 has 3 significant figures, and the density is reported to 3 significant figures. So the rule holds true.)
2. **Addition or Subtraction:** The final result should retain as many decimal places as the number with the *least* decimal places. (e.g., $436.32 \text{ g} + 227.2 \text{ g} + 0.301 \text{ g} = 663.8 \text{ g}$, rounded to one decimal place because 227.2 g has one decimal place) .

Rounding off Uncertain Digits :

- If the insignificant digit to be dropped is **more than 5**, the preceding digit is raised by 1.
- If the insignificant digit to be dropped is **less than 5**, the preceding digit is left unchanged.
- If the insignificant digit to be dropped is **exactly 5**:
 - If the preceding digit is **even**, the insignificant digit is simply dropped (e.g., 2.745 rounded to three significant figures becomes 2.74).
 - If the preceding digit is **odd**, the preceding digit is raised by 1 (e.g., 2.735 rounded to three significant figures becomes 2.74).

In complex multi-step calculations, retain one digit more than the significant digits in intermediate steps, and round off at the end. Exact numbers (like 2 in

$r = d/2$) have an infinite number of significant figures.

1.4 Dimensions of Physical Quantities

The

dimensions describe the fundamental nature of a physical quantity. All derived physical quantities can be expressed in terms of combinations of the seven fundamental base quantities, referred to as the seven dimensions of the physical world: [L] (length), [M] (mass), [T] (time), [A] (electric current), [K] (thermodynamic temperature), [cd] (luminous intensity), and [mol] (amount of substance). The dimensions of a physical quantity are the powers to which the base quantities are raised to represent that quantity. For instance, the dimensions of volume are

$[L^3]$ (or $[M^0 L^3 T^0]$), and force is

$[MLT^{-2}]$. Magnitudes are not considered in dimensional representation; it's about the type of physical quantity.

1.5 Dimensional Formulae and Dimensional Equations

- **Dimensional Formula:** An expression that shows how and which base quantities represent the dimensions of a physical quantity.
 - Volume:

$$[M^0 L^3 T^0]$$
 - Speed/Velocity:

$$[M^0 L T^{-1}]$$
 - Acceleration:

$$[M^0 L T^{-2}]$$
 - Mass Density:

$$[M L^{-3} T^0]$$
- **Dimensional Equation:** An equation formed by equating a physical quantity with its dimensional formula.
 - $[V] = [M^0 L^3 T^0]$
 - $[v] = [M^0 L T^{-1}]$
 - $[F] = [M L T^{-2}]$
 - $[\rho] = [M L^{-3} T^0]$

1.6 Dimensional Analysis and its Applications

Dimensional analysis is crucial for describing physical behavior because only quantities with the same dimensions can be added or subtracted. Units and dimensions can be treated like algebraic symbols in multiplication and division. Both sides of a mathematical equation must have the same dimensions.

1.6.1 Checking the Dimensional Consistency of Equations:

This principle, called the

principle of homogeneity of dimensions, states that if the dimensions of all terms in an equation are not the same, the equation is incorrect. For example, in the equation

$$x = x_0 + v_0 t + \frac{1}{2} a t^2:$$

- $[x] = [L]$
- $[x_0] = [L]$
- $[v_0 t] = [L T^{-1}][T] = [L]$
- $[(1/2) a t^2] = [L T^{-2}][T^2] = [L]$

Since all terms have the dimension of length, the equation is dimensionally correct.

While dimensional consistency is a preliminary test, it doesn't guarantee a correct equation (due to dimensionless constants or functions). However, a dimensionally inconsistent equation is definitively wrong. Arguments of special functions (trigonometric, logarithmic, exponential) must be dimensionless. Pure numbers or ratios of similar physical quantities (like angle or refractive index) have no dimensions.

1.6.2 Deducing Relation among the Physical Quantities:

The method of dimensions can be used to derive relationships between physical quantities if their dependencies are known (up to three physical quantities or linearly independent variables). This involves expressing the dependence as a product type with unknown exponents and then equating the dimensions on both sides of the equation.

For example, to derive the time period (T) of a simple pendulum dependent on length (l), mass (m), and acceleration due to gravity (g) :

$$T = k l^x g^y m^z$$

By equating dimensions:

$$[L \cdot M \cdot T^1] = [L^1]^x [L^1 T^{-2}]^y [M^1]^z = L^x T^{-2y} M^z$$

This leads to:

$$x + y = 0, -2y = 1, \text{ and } z = 0.$$

Solving these equations gives

$$x=1/2, y=-1/2, z=0.$$

Thus,

$$T=kl^{1/2}g^{-1/2} \text{ or } T=kl/g \quad .$$

The value of the dimensionless constant 'k' (e.g.,

2π for a simple pendulum) cannot be determined by dimensional analysis. This method tests dimensional validity but not the exact relationship and cannot distinguish between physical quantities having the same dimensions.